Dominion Energy Virginia reliability: what public data shows, and where I'd look first

Independent analysis of public data

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Data: EIA-861 · PUDL · NOAA Storm Events · EAGLE-I · EIA-930

Outage minutes rose from 130 to 133 because interruptions became more frequent, not longer

130 → 133

Reported 3-year average outage minutes per customer, ex-major events (2022–24 → 2023–25)

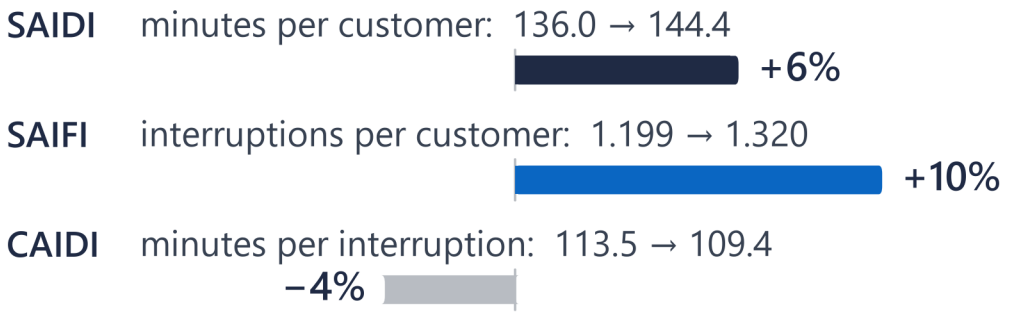
130.2 → 133.0

The same figures rebuilt from public EIA-861 data: they reconcile

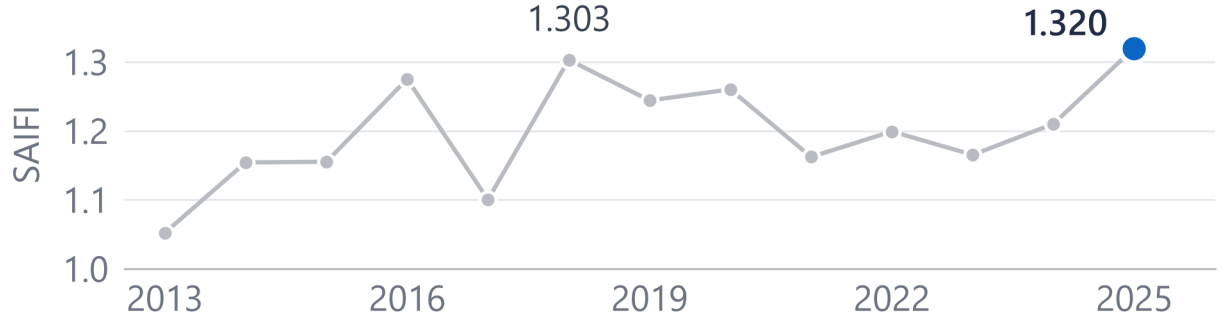
136.0 → 144.4

The average moved because 2025 replaced 2022 in the 3-year window

2022 → 2025, the two years that move the 3-year average



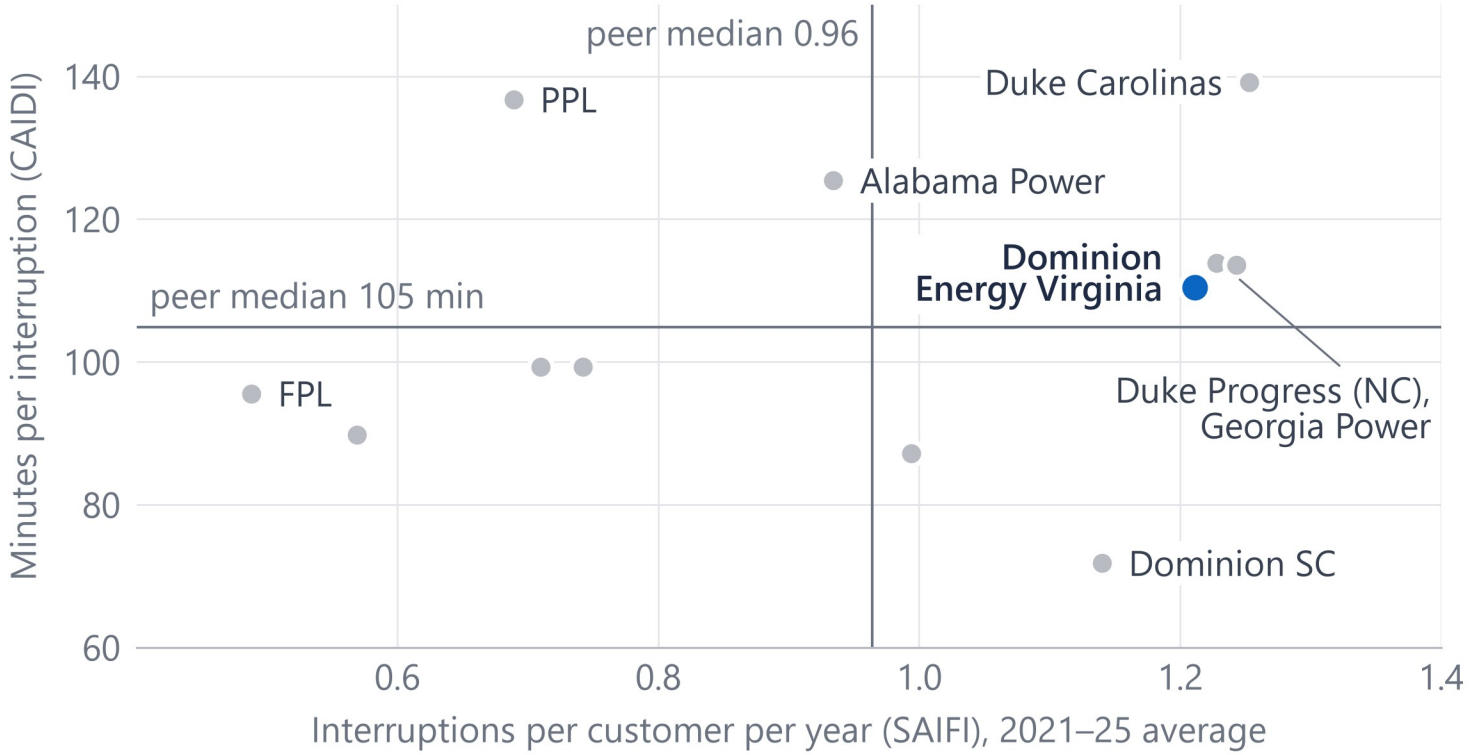
SAIFI by year: 2025 is the highest of 2013–2025



For Dominion: the first question is what is making interruptions more frequent, before how to restore faster.

Source: EIA-861 2025 final and PUDL (EIA-861) 2013–2024, Dominion Energy Virginia, excluding major event days; 3-year means of annual values. Reported figures: Dominion Energy 2025 Annual Report, p. 10.

Among 12 peer utilities, Dominion restores at a mid-pack pace; frequency is the gap



1.21

interruptions per customer a year: 4th highest of 12 (peer median 0.96)

110 min

per interruption: 6th of 12 (peer median 105)

Methods differ: Dominion uses its own major-event rule, most peers IEEE 1366, so ranks are indicative.

For Dominion: preventing interruptions is where the gap to peers is; restoration speed is already near the middle.

Source: EIA-861 / PUDL 2021–2025, excluding major event days, means of annual values; twelve large investor-owned utilities in PJM and the Southeast, selected by EIA ID before results were computed.

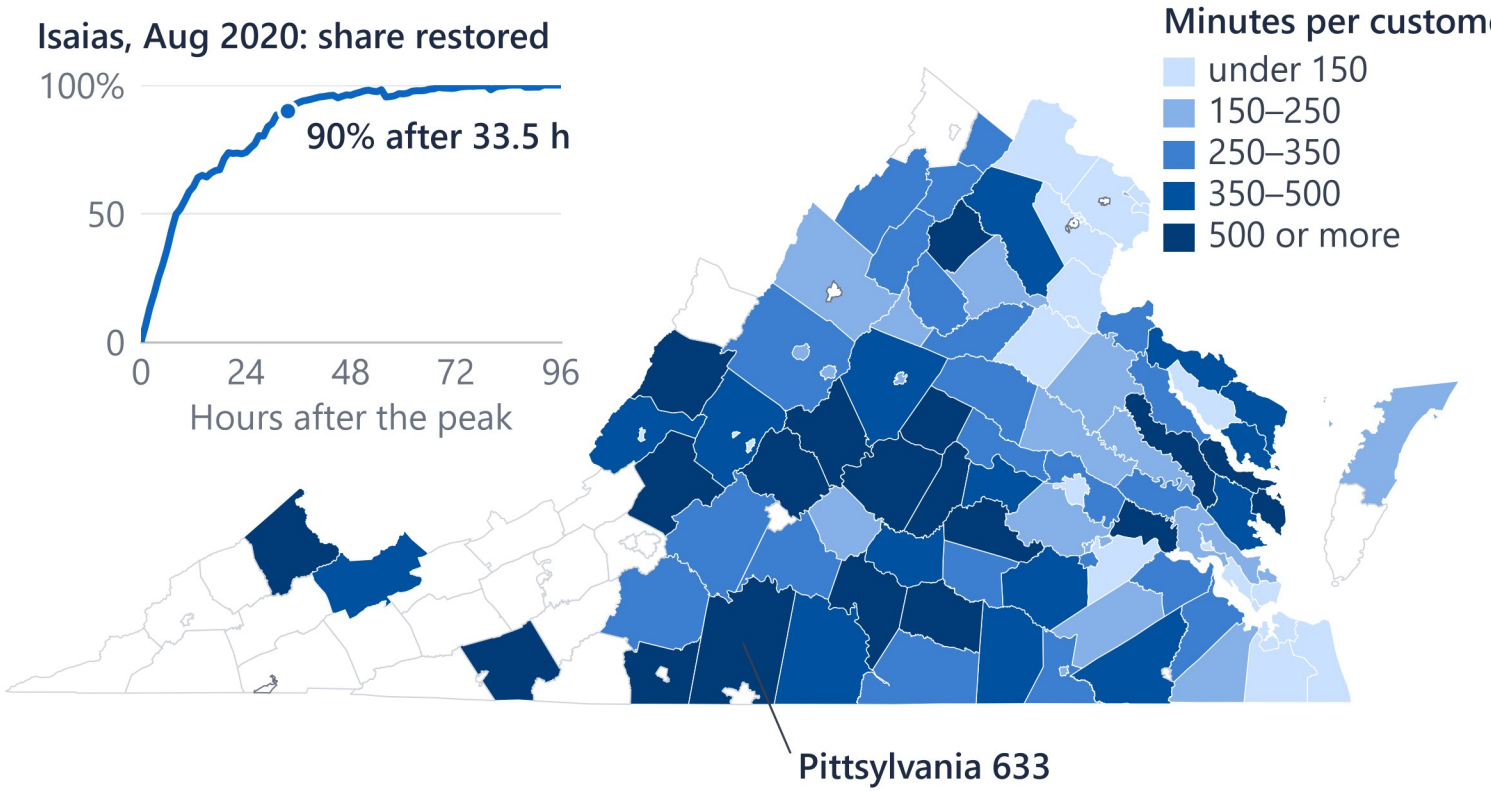
County-days with a storm report had about 12 times the outage customer-hours of calm days

11.9×

outage customer-hours per 1,000 customers vs calm county-days (95% CI 7.9–16.9)

Holds on non-major days (9.9×), in Dominion-only counties (19.9×) and in every year

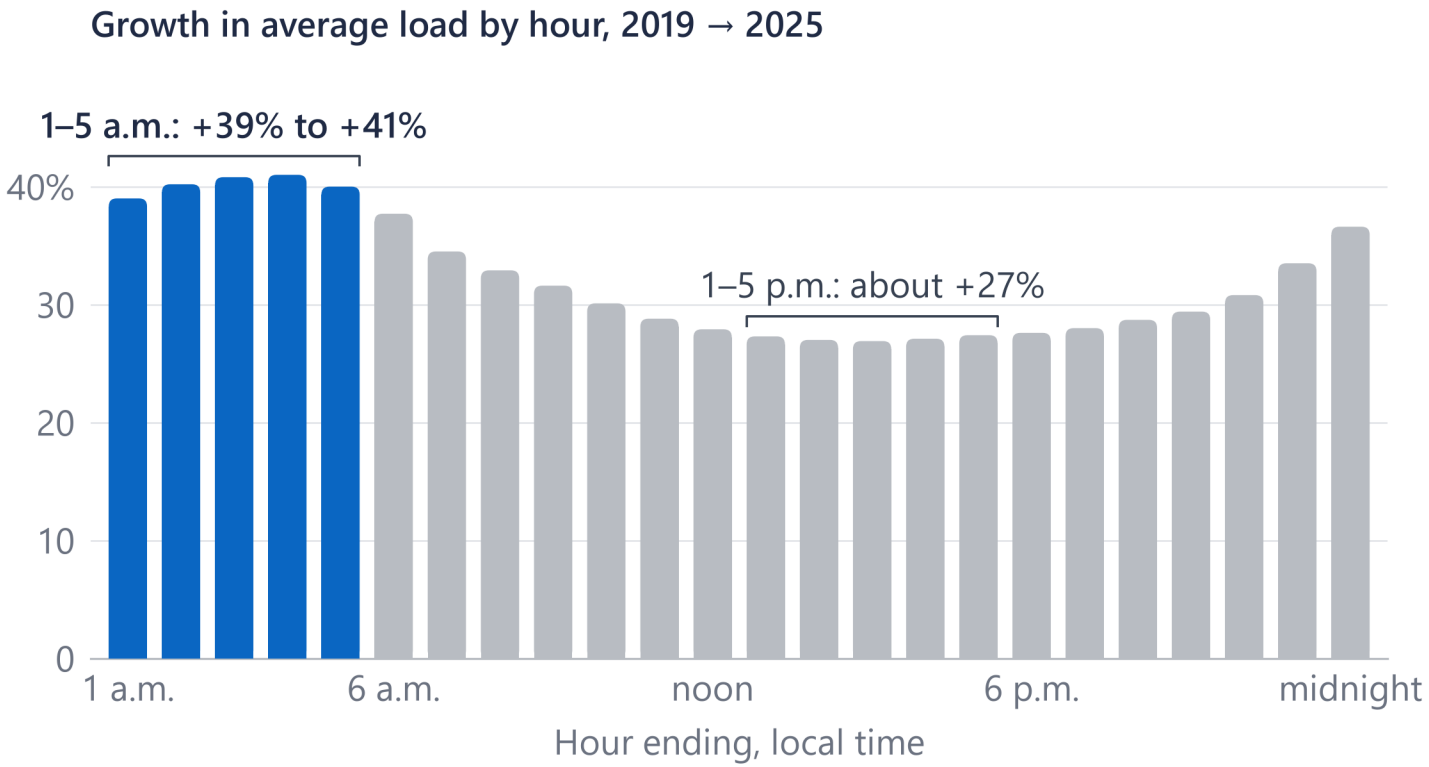
EAGLE-I counts every utility in a county. 6,108 storm county-days, 2015–2025.



For Dominion: read each year against its storms, and start with the counties that lose the most minutes on ordinary days.

Source: EAGLE-I (ORNL) county outages, all utilities, 99 of Dominion's 103 EIA-861 service-territory counties (4 cities left off: unusable data); NOAA Storm Events. Map: non-major-day outage minutes per customer, 2015–24 average.

Overnight demand grew faster than afternoon demand, and winter rivals summer for the yearly peak



+39–41%

average load, hours ending 1–5 a.m., 2019 → 2025 (1–5 p.m.: about +27%)

Winter

set the yearly peak in 2019, 2022 and 2025; 2026 to 29 Oct: 25,413 vs 25,384 MW

Peak hour +26%, lowest hour +52%.

For Dominion: less overnight slack and winter peaks: plans and reports built around summer afternoons need updating.

Source: EIA-930 hourly demand, PJM Dominion (DOM) zone, which is wider than Dominion's retail territory; 2019 and 2025 full years; 2026 to the hour ending midnight, 29 Oct. Winter = Dec–Feb, summer = Jun–Aug.

With Dominion's outage, meter and circuit data, I would start where public data stops

Days 1–30

Reconcile

Learn how outages, meter events and circuits are recorded and reported.

Rebuild one year's SAIDI, SAIFI and CAIDI and reconcile them to the filed figures.

Deliverable: a reconciled baseline with written definitions

Days 31–60

Diagnose

Interruptions by circuit and cause code, vegetation included; repeat interruptions (CEMI).

Restoration timed from meter “last gasp” (99.8% of meters are smart).

Deliverable: tested views and a data dictionary

Days 61–90

Prioritise and track

Rank feeders by customer-minutes; track Grid Resiliency work such as mainfeeder hardening against similar untreated feeders and weather.

Deliverable: a Power BI programme-tracking report, refreshed on a schedule

For Dominion: each step turns a system-level trend into a circuit-level view the programme team can act on.

Source: A plan: Dominion data is named by type only, and none was used here. Smart meters: EIA-861 2025 (AMI share of meters). Grid Resiliency: Dominion Energy website, read 1 November 2026.

Built like governed production reporting: tested, documented, and reconciled to the reported numbers



12 tests pass

keys, SAIDI = SAIFI × CAIDI, EIA's published 2025 values

Data dictionary

a business definition for every column and measure

Decisions log

22 decisions, such as storm types and peers; peers fixed before any results

Reconciled

12 of 12 rebuilt values match the published figures, such as 133.0 vs 133

For Dominion: reconcile first, test every load, document every decision: the same discipline for Dominion's reporting.

Source: Project build: SQL scripts, tests T1–T12, data dictionary and decisions log D1–D22; row counts as loaded.

Three takeaways: frequency first, reliability net of weather, plan for a new load shape

1 **Look at frequency first**

Interruptions per customer +10% (2022→2025), the 2013–2025 high; 4th of 12 peers

2 **Net out the weather**

12× the outage customer-hours on storm county-days as on calm ones: an association

3 **Plan for a new load shape**

Hours ending 1–5 a.m. +39–41% (2019→2025); winter set the peak in 2019, 2022 and 2025

I build governed Snowflake → Power BI reporting, and I've already started learning your reliability data.

How my experience maps — Mercury Insurance, Data Analyst

Snowflake pipeline, 20M+ rows	Root-cause analysis on joined data	Validation and reconciliation	Metadata and business definitions	Requirements with operations and IT
→ this mart	→ SAIFI by cause	→ reconciled 133.0	→ data dictionary	→ programme teams

For Dominion: I would like to do this with your data.

Source: Numbers from slides 2–5. Experience: Mercury Insurance, Data Analyst, 2026 to present.

Definitions: SAIDI = SAIFI × CAIDI, and major event days are set aside by rule

SAIDI
outage minutes per customer per year

=

SAIFI
interruptions per customer per year: how often

×

CAIDI
minutes per interruption: how long

Dominion 2025: 1.320 × 109.4 = 144.4 minutes (excluding major event days)

Major event days (MED) are set aside so that one extreme storm does not swamp the trend. IEEE 1366 marks a day as a major event day when its SAIDI exceeds a threshold set from the previous five years of daily SAIDI (the “2.5 beta” method).

Dominion reports to EIA under its own major-event rule; most peers in this deck use IEEE 1366. Headline figures here exclude major event days; all-event figures are on A3.

Log split: ln(SAIDI ratio) = ln(SAIFI ratio) + ln(CAIDI ratio). Share from frequency = ln(SAIFI ratio) ÷ ln(SAIDI ratio): 2023→2025, 0.125 ÷ 0.158 = 79%.

Source: IEEE Std 1366 (reliability indices); EIA-861 reliability, Dominion Energy Virginia 2023 and 2025.

The weather fit is a check on one year, not a forecast

Year	Storm county-days	SAIDI ex-MED	From the fit	Difference
2015	521	119.6	131.3	−11.6
2016	594	138.2	135.0	+3.2
2017	419	117.4	126.0	−8.6
2018	683	149.6	139.7	+9.9
2019	848	149.9	148.2	+1.7
2020	429	141.0	126.5	+14.5
2021	673	133.6	139.1	−5.5
2022	718	136.0	141.5	−5.4
2023	390	123.3	124.5	−1.1
2024	464	131.3	128.3	+3.0
2025 (not in fit)	**791**	**144.4**	**145.3**	**−0.9**

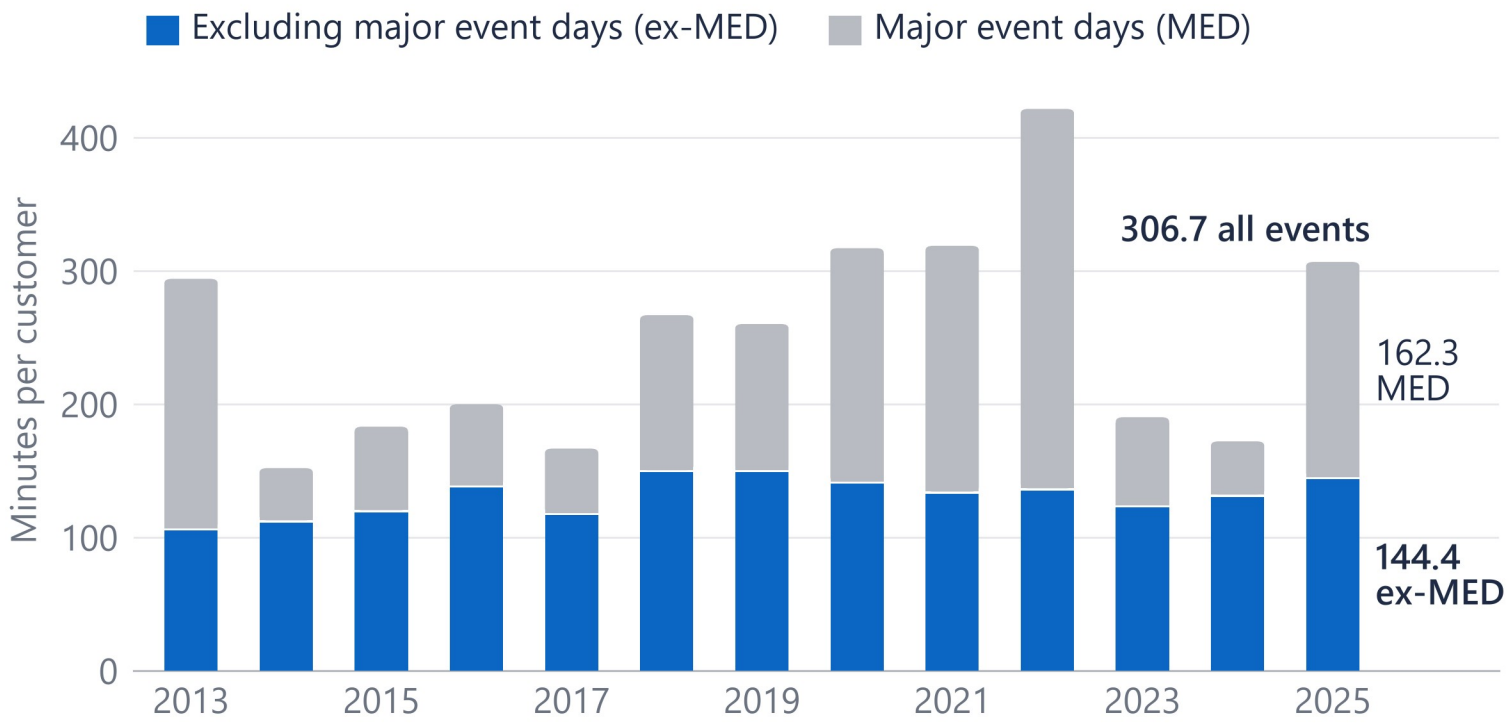
Fit on 2015–2024 (n = 10): SAIDI = 104.3 + 0.0518 × storm county-days; r = 0.70 (t = 2.75, p = 0.025; 95% CI 0.12 to 0.92), R^2 0.49.

2025 was left out: the fit gives 145.3, the actual was 144.4.

Adjusted for storm days, no clear trend since 2015: about +0.3 minutes a year (95% range −1.4 to +2.1).

Source: NOAA Storm Events and EIA-861, 2015–2025; ordinary least squares on ten annual points (exploratory).

Major event days were about half of all-event outage minutes in 2025



53%

of 2025's all-event outage minutes fell on major event days (162.3 of 306.7)

Major-event minutes swing most: 285 in 2022, 41 in 2024.

Major-event minutes are EIA's all-events SAIDI minus ex-MED SAIDI, not a storm attribution.

Source: EIA-861 / PUDL 2013–2025, Dominion Energy Virginia: SAIDI with and without major event days.

Every number traces to a public source and a tested query

Source	Publisher	Years	Grain	Rows
EIA-861 Reliability, 2025 final	EIA	2025	utility × state × year	977
PUDL EIA-861 reliability	Catalyst Cooperative	2013–2024	utility × state × year	24,358
EIA-861 Service Territory	EIA	2025	utility × county	11,803
NOAA Storm Events	NOAA NCEI	2013–2025	storm event	831,288
EAGLE-I county outages	ORNL for DOE	Nov 2014–2025	county × 15 minutes	11,608,442
EIA-930 hourly demand	EIA	Jul 2018–Oct 2026	region × hour	6,077,232
Dominion Energy 2025 Annual Report	Dominion Energy	2025	document (130, 133)	reference

Tests after each load (T1–T12, all pass): no duplicate keys; SAIDI = SAIFI × CAIDI within 2%; Dominion 2025 equals EIA's published 306.7, 144.4 and 2,675,735 customers; every county mapped; full years; no duplicate hours. **T13–T31** check the county outage, storm and load tables (all pass).

Source: Project documentation: sources register (publisher, vintage, licence) and data-quality tests; all public data.

What public data cannot show, and what would answer it

Public data limit	What would answer it (Dominion data, by type)
One number per utility per year (EIA-861)	Outage records by event, circuit and cause
Storms counted by county, not by line	Outage records with cause codes and circuit location
Methods differ between utilities	Dominion's daily SAIDI and its filing method
The DOM zone is wider than the retail territory	Meter interval data by circuit
No customer identities: growth can't be attributed	Customer class and large-load records

Source: Project scope limits (decision D14) and the sources above.

The ten largest outage events since 2014: three tropical, two winter, five wind

#	Peak day	Storm type	Customers out at the peak	Hours to 50% restored	Hours to 90% restored	Customer-hours (m)
1	12 Oct 2018	Tropical (Michael)	558,729	15.8	60.5	17.8
2	2 Mar 2018	Wind	419,136	18.8	55.0	13.2
3	**4 Aug 2020**	**Tropical (Isaias)**	**351,738**	**8.3**	**33.5**	**6.2**
4	3 Jan 2022	Winter	314,644	21.3	102.8	13.7
5	13 Feb 2021	Winter	281,707	40.5	101.5	16.5
6	9 Oct 2016	Tropical (Matthew)	269,298	11.3	52.8	8.0
7	16 Jun 2016	Wind	234,951	19.5	65.8	7.0
8	19 Jun 2025	Wind	225,609	4.5	22.0	2.1
9	16 Feb 2025	Wind	187,074	3.0	20.3	8.2
10	14 May 2018	Wind	159,532	4.5	20.0	1.4

Source: EAGLE-I (ORNL), all utilities' customers in Dominion's 103 counties, Nov 2014–2025; storm type from NOAA Storm Events and HURDAT2. Hours from the peak. The Feb 2025 event merges two storms.